

Otorhinolaryngology & Head-Neck Surgery

Departmental Objectives

The aim is to teach undergraduate medical students so as to produce need based community oriented doctors who will be capable of :

1. diagnosing and managing common ENT & Head-Neck disorders.
2. referring complicated ENT and head-neck disorders to appropriate centres if and when necessary
3. managing common emergencies in ENT & head-neck disease
4. giving preventive advice on certain aspects of ENT & head-neck diseases

To achieve above mentioned departmental objectives the following learning objectives should be achieved:

1. The art of appropriate history taking
2. Should perform primary ENT & head-neck examination procedure
3. Should use the aural speculum, nasal speculum, tongue depressor, laryngeal mirror, tuning fork and head mirror/light, otoscope & other instruments as listed in the enclosure
4. Should be able to describe the clinical application of basic anatomy & physiology of Ear, Nose and Throat
5. Should be able to describe the pathology of common ENT disorders & disorders of the Head-Neck region
6. Should list commonly used drugs and describe their adverse effects
7. Should recommend common investigative procedures and special investigation (CT, MRI, and sonography, etc)

Learning Objectives and Course Contents in Otorhinolaryngology & Head-Neck Surgery

Learning Objectives	Contents	Teaching Hours
Students will be able to: 1. demonstrate the applied Anatomy of ear. 2. demonstrate the applied Physiology of ear. 3. take History of ear diseases 4. conduct clinical hearing test and value the significance of audiometry and caloric test. 5. diagnose various ear diseases by clinical examination (FB, Otitis Externa, Traumatic Tympanic membrane perforation, ASOM, CSOM, Otosclerosis. 6. remove impacted wax, foreign body, Aural toileting 7. diagnose ear diseases and Its complications and refer to appropriate hospital when needed. e.g.- perichondritis otосclerosis extra and intracranial complications of middle ear diseases 8. make D/D of earache 9. differentiate safe from unsafe variety of CSOM.	<u>EAR</u> <u>CORE</u> 1. applied Anatomy of ear 2. applied Physiology of ear:- hearing, Balance 3. congenital diseases of ear-Preauricular sinus 4. causes of earache 5. causes of deafness 6. diseases of ext. ear-Furuncle, Otitis externa ,Otomycosis, Foreign body, Trauma,Perichondritis of pinna 7. diseases of middle ear-ASOM, CSOM, OME, Otosclerosis. 8. diseases of internal Ear-Meniere's disease, Labyrinthitis. 9. Tuning fork test, Audio metry, Caloric test 10. micro ear surgery-Myringotomy Myingoplasty & different types of mastoidectomies. 11. neurootological complications: Lateral sinus thrombosis, general idea about intra cranial complications of ASOM & CSOM. <u>Additional:</u> 12. causes of Vertigo &Tinnitus 13. management of deafness.	

Learning Objectives	Contents	Teaching Hours
Student will be able to : 1. describe applied anatomy and applied physiology of nose. 2. manage epistaxis 3. remove FB and reduction of Fracture nasal bone. 4. diagnose nasal diseases by clinical examinations 5. refer the patient to specialized ENT centre 6. apply ANS Pack. 7. history taking of disease of Nose and PNS.	<u>NOSE</u> CORE: 1. Anatomy of nose 2. Physiology of nose 3. Epistaxis. 4. FB nose, Fracture nasal bone 5. Nasal allergy 6. Nasal polyp 7. Rhinitis, Sinusitis 8. DNS, septal perforation, septal abscess, septal haematoma 9. Nasal papilloma, rhinosporidiosis. 10. Atrophic rhinitis 11. Nasopharyngeal angiofibroma and naso-pharyngeal carcinoma. 12. Sino-nasal malignancy Additional Headache Tumours of nose and PNS Common nasal and sinus Operation:- Polypectomy SMR, Septoplasty Caldwell Luc operation BAWO	

Learning Objectives	Contents	Teaching Hours
Student will be able to : 1. Describe anatomy of oral cavity, pharynx, larynx and oesophagus. 2. Describe Physiology of deglutition. 3. Make D/D of white patches, ulcers in oral cavity, Leukoplakia and Sorethroat. 4. Diagnose Diphtheria and refer it to appropriate hospital 5. Diagnose acute & recurrent tonsillitis, adenoids, 6. Describe indications of adenotonsillectomy and principles of post operative management and contraindications. 7. Diagnose complications of adenotonsillectomy and its management 8. List D/D of dysphagia. 9. List D/D of hoarseness of Voice. 10. List D/D of Stridor 11. Describe indications of tracheostomy & its steps, postoperative management and complications.	Mouth cavity, pharynx, larynx and esophagus CORE 1. Anatomy of oral cavity, pharynx, larynx and Oesophagus 2. Physiology of salivation, deglutition and functions of larynx, pharynx. 3. Diseases of oral cavity Congenital anomalies like Hare lip, cleft palate White patch-oral cavity, oral ulceration, Leukoplakia and neoplasm. 4. Acute & recurrent tonsillitis faucial diphtheria. 5. Adenoids 6. Tonsillectomy and adenoidectomy 7. Peritonsillar abscess, retro pharyngeal abscess, parapharyngeal abscess. Larynx Acute Epiglottitis, Acute Laryngo tracheo bronchitis Acute & chronic laryngitis Papillomalarynx Stridor Causes of hoarseness of voice Tracheostomy Carcinoma-larynx. Foreign Body larynx, trachea, bronchus.	

Learning Objectives	Contents	Teaching Hours
	<u>Pharynx</u> FB Malignancy of Pharynx <u>Oesophagus</u> PV syndrome Dysphagia Foreign Body Benign & malignant lesion of Oesophagus (strictures, rupture) <u>Head-Neck</u> 1. Applied anatomy of salivary glands, Thyroid & Parathyroid glands 2. Physiology of salivary glands, Thyroid & Parathyroid glands 3. Salivary gland diseases 4. Thyroid and parathyroid diseases 5. Neck mass 6. Congenital sinus & cyst of head neck (Thyroglossal cyst, Branchial cyst, Branchial sinus) <u>General Idea about head neck malignancies</u>	

Integrated Teaching

Topic	Learning Objective	Teaching Aids	Assessment	Department
Otogenic and Rhinogenic extra-cranial & intra-cranial complications	Student will be able to: - state the causes of extra-cranial & intra-cranial complications of ASOM and CSOM - describe the symptoms & signs of acute mastoiditis, facial palsy, labyrinthitis lateral sinus thrombosis. - Investigate & interpret the results of investigation. - treat different complications (gross idea)	Video cassette film of C.T. Scan, X-ray, Diagram, Otoscope, Hammer, Cotton, Pin & Patients. Tongue depressor, PNS mirror, laryngeal mirror Nasal speculum. (Nice to know fundoscopy) Ophthalmoscope	Performance, Interpretation, Short Question, Modified short Question, MCQ Practical Exam OSCE	ENT & Neuro Surgery ENT & Eye
Facio-Maxillary Neoplasm	State common causes of maxillary swelling/carcinoma of Maxilla.			

Teaching Methods:

- Lecture/ Mini Lecture
- Tutorial/ Demonstration - Video
- Case presentation- Subject – Operation- Programe side Teaching Theatres
- Discussion, Visit to RHC / Specialised Centre (If available)

Teaching Hours for Otorhinolaryngology & Head-Neck Surgery

Methods	Total
Lectures	38 hours
Ward Teaching	8 weeks

Instruments

1. Ear speculum
2. Otoscope
3. Nasal speculum (Thudicum)
4. Antrum puncture trocar and cannula (Lichwitz)
5. Tongue depressor (Luc's)
6. PNS mirror
7. Laryngeal mirror
8. Boyle Davis mouth gag
9. Adenoid curette with / without cage (St Clare Thomson)
10. Tracheostomy tube-metallic/PVC
11. Laryngoscope
12. Oesophagoscope
13. Bronchoscope
14. Head light/mirror
15. Tuning Fork

Operative Procedures

- a. Tonsillectomy
- b. Adenoidectomy
- c. Septoplasty/SMR
- d. Caldwell-Luc operation
- e. Myringoplasty
- f. Mastoidectomy
- g. Thyroidectomy
- h. Salivary gland excision
- i. Biopsy for diagnosis of carcinoma of tongue, oral lesions etc
- j. Direct laryngoscopy
- k. Neck node biopsy
- l. Antral washout

X-ray

- m. X-ray paranasal sinus (occipito-mental view)
- n. X-ray nasopharynx – lateral view
- o. X-ray mastoid
 - Towne's view
- p. X-ray neck
 - Lateral view
 - Ba swallow x-ray of esophagus

Nice to know

CT scan /MRI

FOL – Fibre Optic Laryngoscopy

CLINICAL PLACEMENT OF STUDENTS DURING PHASE II, III & IV (for 62 weeks)

WEEKS	PHASE II 20 WEEKS	WEEKS	PHASE III 14 WEEKS	WEEKS	PHASE IV 12+12 + 04 WEEKS
01-15	Surgery indoor Surgery OPD	01-04	Orthopaedics & traumatology		1 st term
16-17	Orthopaedic surgery	05-08	Ophthalmology	01-04	Orthopedics
18	Radiology	09-12	ENT	05-08	Ophthalmology
19	Anaesthesia	13	Radiotherapy	09-12	ENT
20	Dentistry	14	Neurosurgery		2 nd & final term
		15	General Surgery		
Card completion exam at the end of rotation & Term exam at 41 st week		Term exam at 41 st week		01-07	Surgery
				08-09	Urology
				10-11	Paediatric Surgery
				12	Emergency & Casualty Burn & Plastic Surgery
				04 weeks	BLOCK POSTING
					Final assessment
Time schedule for the lecture classes (number)					
DICIPLINE		2ND PHASE	3RD PHASE	4TH PHASE	TOTAL
Gen Surgery		35	30	60	125
Orthosurgery		0	15	45	60
Radiology		0	0	5	5
Radiotherapy		0	0	8	8
Anaesthesia		0	10	0	10
Neurosurgery		0	0	5	5
Paediactric Surgery		0	5	10	15
Urology		0	5	10	15
Burn Plastic Surgery		0	0	5	5
		35	65	148	248

Large Group Teaching

All lectures should be interactive one.

It should be directed to develop analytical and problem solving attitude.

Student should be encouraged to adopt self-directed learning.

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Clinical Teaching and tutorials

- Students are to attend the wards as per placements twice in a day in morning and evening sessions as fixed by the respective college authority.
- They must be well dressed along with apron & nameplate. Shirts, Pants (full length) Shoes only and Winter apparels are allowed in ward settings. Three quarter pant, T-shirt, Sandals are not allowed and teacher may disallow those students to continue the class.
- They will carry stethoscope, percussion hammer, pencil torch and measuring tape and other necessary clinical examination tools.
- During their ward visit, they will examine patients and try to make working diagnosis and write the history as per prescribed format.
- They will go through hospital documents and look what necessary measures and decision has been taken to follow the management of the patient in the ward.
- They will observe and practice techniques of IV & IM injection, infusion, dressing of the wounds. Student will also attend the operation theater and observe the instruments and equipments used in the operation theater.
- They will observe the techniques of different anaesthesia and the drugs used, techniques of hand scrubbing, gowning gloving, scrubbing and draping of operation field, making incisions, haemostasis, saturating technique and wound repair.
- Students performance will be assessed by item examinations, ward and term examinations.

Assessment-

1. Internal assessment: (Marks for formative assessment)

- a. Items & Card completion examination ,
- b. Year final assessment at the end of Phase-II & III (written)
- c. MCQ in Integrated teaching.
- d. Final assessment examination (similar to final professional examination) at the end of block posting. (Medicine, Surgery, Obs & Gynae)

2. Final professional MBBS Examination:

- a. **Written:** (MCQ- 20 (10 SBA and 10 MTF) ; SAQ & SEQ=70) each paper
Time allocation: MCQ- 30 minutes; SAQ+SEQ – 02hrs 30minutes.
 - i. Paper – I SAQ & SEQ consists of 4 groups.

Group -1:- Principles of surgery, Vascular Diseases, Anaesthesia, Radiology, Radiotherapy.
 Group -2:- GIT, Paediatric surgery, Operative Surgery, Chest disease
 Group -3:- HBS & Pancreas, Urology, Breast, Endocrine.
 Group -4:- Orthopaedics & Traumatology, Neurosurgery,
 There will be 05 questions in each group and out of those 04 are to be answered carrying 3.5 marks each.
 At least two Structured Essay Question (SEQ) will be in each paper.

- ii. Paper –II:
 Ophthalmology-- MCQ-10 (5 SBA, 5 MTF) & SAQ & SEQ -35;
 ENT-- MCQ-10 (5 SBA, 5 MTF) & SAQ & SEQ -35
 Group-1and Group 2 = Ophthalmology
 Group-3 and Group 4 = ENT
 At least two Structured Essay Question (SEQ) will be in each paper.
- iii Preferably questions will be of
 recall type- 30%,
 understanding or data interpretation type- 30% and
 problem solving type- 40%
- iv Question should cover the whole curriculum .
 90% of the questions should preferably be from core content
 and 10% from additional content of course.
- v. **Scripts distribution:** Group-1 will be assessed by General surgeon, Group -2 will be assessed by General surgeon, Group-3 will be assessed by a General surgeon/allied subject expert and Group -4 will be assessed by an Orthopedic surgeon.

b. OSPE –

- i. Stations will be constructed centrally by two experience examiners nominated and supervised by chairman of the examination committee of the respective university.
- ii. Each station will of 5 minutes time and marks will be allocated according to rules mentioned in the subject concerned.
- iii. All the examinee under each university will appear in OSCE/OSPE exam in their designated centers on a same date and before 9 am scheduled by University for a particular subject. Failure to arrive at examination center before 9 am is an offense and examiner may dis qualify the candidate.
- iv. OSPE examination of Surgery, Ophthalmology and ENT will be in two different days.
- v. Answer scripts of OSPE will be divided among the examiners for evaluation and the marks are to be submitted prior to final day of the oral examination scheduled in the respective centre.
- vi. Every examination center should be prepared for testing competencies including different procedure stations, data analysis, counseling, displaying x-ray, specimens and instruments. Original materials should be placed at each station.
- vii. Station setup
 - 1. Total 20 stations will be made comprising 10 from Surgery,
 - 2. Five (5) Ophthalmology and
 - 3. Five (5) ENT stations.

Out of those, at least two stations from surgery, one from Ophthalmology and one from ENT will be procedural station.

ix Marks allocation

Surgical stations are- (10 x 6 = 60 marks)

	No.
a. Plain x-ray	-1
b. Contrast x-ray	-1
c. Orthopaedic X-ray	-1
d. Specimen	-1
e. Instrument/s	-1
f. Appliances (Catheter, tubes, stoma or reservoir bags etc)	-1
g. Data interpretation	-1
h. Procedure stations	-2
i. Splint/bandage	-1
4. Ophthalmology -5 and ENT-5 stations are- (5+5) x4 = 40 marks)	
	No.
a. Instrument station	= 1
b. X-ray station/ Specimen	= 1
c. Clinical photograph/ tracing	= 1
d. Procedure	= 1

c. Structured Oral Examination. (SOE)

NB: Oral & Practical Examination Boards of **Surgery & Allied Subjects: Eight (8) Examiners in 4 boards in two days.**

Day -1:

Board- A- 1 examiner from General Surgery & 1 examiner from Allied subjects

Board-B-1 examiner from General Surgery & 1 examiner from Orthopaedics

Day-2:

Board-A-1 examiner from Ophthalmology & 1 examiner from Ophthalmology

Board-B-1 examiner from ENT & 1 examiner from ENT

NB: In case of unavailability of any concerned examiner of any board the convener of the examiner in consultation with concerned dean of the faculty of medicine will select the examiner from General surgery or sub specialty or any allied subject

Paper-1 (General surgery and allied subjects)

Marks-30X2=60

- Two separate boards comprising one internal and one external examiner will assess written scripts, oral, practical and clinical examination.
- There are two other reserve examiners in each internal and external pool. One of the reserve examiner should be from allied subject like urology, pediatric surgery, plastic surgery or neurosurgery.
- Out of four examiners two will be from general surgery, one will be an orthopedician & another one will be from allied subjects of surgery.
- There will be four boxes covering questions on surgery and allied specialties assigned for each examiner.
- Each box will contain at least 20 sets of questions.
- A set of question will contain 3 small questions of three-difficulty level (Must Know, Better to Know & Nice to Know)
- Content of the box-
 - Box-1:- Principles of surgery, Vascular Diseases, Anaesthesia, Radiology, Radiotherapy.

2. Box-2:- GIT, Paediatric surgery, Operative Surgery, Chest disease
3. Box-3:- HBS & Pancreas, Urology, Breast, Endocrine.
4. Box-4 :- Orthopaedics & Traumatology, Neurosurgery.

Paper –II (Ophthalmology and ENT)

Marks 20X 2= 40

- i. Two separate boards for each specialty comprising one internal and one external examiners will assess written scripts, oral, practical and clinical examination.
- ii. There will be one reserve examiner in each specialty.
- iii. Instruments and x-rays will not be examined in viva board.
- iv. Each student will be allocated 15 minutes
- v. Problem solving skills / Judgment of knowledge should be examined
- vi. The question and answer will be constructed by the examiners in advance
- vii. Question is typed in a card and put in box of defined domain
- viii. A number of questions from each topic should be constructed covering the content area.
- ix. Content will be changed on alternate days
- x. The candidate randomly selects one card from each box and answer.
- xi. The candidate should answer selected number of question in the board
- xii. The examiner read the question, repeat it if necessary or the candidate reads the question if allowed.
- xiii. When candidate answers the questions, the examiner will put a tick in appropriate site on a prepared rating scale

d. Clinical examination of surgery

- a. Surgery -60
 1. Short cases 3 x 10 =30
 2. One Long case- 30.
- b. Ophthalmology cases -2 x 10=20
- c. ENT cases- 2 x 10=20

Mark distribution of oral, clinical and practical examination in surgery in final professional examination

Subject	Oral	Practical/OSPE	Clinical	Total
Surgery + Allied & Orthopaedics	30+30	60	30+30	180
Ophthalmology	20	20	20	60
ENT	20	20	20	60
Total	100	100	100	300

- ❑ There will be separate Answer Script for MCQ (SBA & MTF) and written SAQ & SEQ assessment.
- ❑ Pass marks is 60 % in EACH of Written, oral, practical and clinical components.
- ❑ Practical Examination will be in 2 days, one day Surgery, One day Eye-ENT
- ❑ Oral+Clinical will be in 2 days, One day- Surgery, another day- Ophthalmology + ENT.
- ❑ Marks and Written examination scripts must be returned before last day of oral-clinical examination at respective examination center. Otherwise convener of the center will return the whole scripts to Dean office for final decision.
- ❑ For declaration of results in earliest possible time after compilation of marks quick disposal of marks to competent authority is desirable.

FINAL PROFESSIONAL EXAMINATION

Assessment of Surgery (MARKS DISTRIBUTION)

Components	Marks On each component	Sub total Marks	Total Marks
<u>Written examination</u>			
<u>Formative assessment marks</u>			
General Surgery & allied subjects	10		
Ophthalmology	05	20	20
ENT	05		
<u>Written</u>			
Paper – 1:			
General Surgery + allied & Orthopaedics : MCQ- (SBA+MTF) +(SAQ + SEQ)	(20+70)	90	180
Paper – II:			
Ophthalmology: MCQ- (SBA+MTF) +(SAQ + SEQ)	(10+35)	45	
ENT : MCQ- (SBA+MTF) +(SAQ + SEQ)	(10+35)	45	
<u>Oral, Clinical & Practical</u>			
General Surgery + allied & Orthopaedics (Oral+ Clinical+ Practical)	(60+60+60)	180	
Ophthalmology (Oral+ Clinical+ Practical)	(20+20+20)	60	300
ENT (Oral+ Clinical+ Practical)	(20+20+20)	60	
Oral examination should be structured.			
Grand Total Marks			500